The economic burden of dry eye disease in the United States: a decision tree analysis.

PURPOSE: The aim of this study was to estimate both the direct and indirect annual cost of managing dry eye disease (DED) in the United States from a societal and a payer's perspective. METHODS: A decision analytic model was developed to estimate the annual cost for managing a cohort of patients with dry eye with differing severity of symptoms and treatment. The direct costs included ocular lubricants, cyclosporine, punctal plugs, physician visits, and nutritional supplements. The indirect costs were measured as the productivity loss because of absenteeism and presenteeism. The model was populated with data that were obtained from surveys that were completed by dry eye sufferers who were recruited from online databases. Sensitivity analyses were employed to evaluate the impact of changes in parameters on the estimation of costs. All costs were converted to 2008 US dollars. RESULTS: Survey data were collected from 2171 respondents with DED. Our analysis indicated that the average annual cost of managing a patient with dry eye at $783 (variation, $757-$809) from the payers' perspective. When adjusted to the prevalence of DED nationwide, the overall burden of DED for the US healthcare system would be $3.84 billion. From a societal perspective, the average cost of managing DED was estimated to be $11,302 per patient and $55.4 billion to the US society overall. CONCLUSIONS: DED poses a substantial economic burden on the payer and on the society. These findings may provide valuable information for health plans or employers regarding budget estimation.